

Name: _____ Date: _____

Period: _____

Exit Ticket — Answer independently

The table below gives values of a periodic function f over one period.

x	0	2	4	6	8	10
$f(x)$	3	11	3	-5	3	11

1. Identify the maximum and minimum values of f .

Maximum = _____ Minimum = _____

2. Compute the amplitude and midline.

$$A = \frac{\max - \min}{2} = \underline{\hspace{2cm}}$$

$$D = \frac{\max + \min}{2} = \underline{\hspace{2cm}}$$

3. Estimate the period k from the table. Explain how you found it.

$k = \underline{\hspace{2cm}}$ Explanation: _____

4. Using your period, compute $B = \frac{2\pi}{k} = \underline{\hspace{2cm}}$

5. A student uses the model $f(x) = 8\cos\left(\frac{\pi}{4}(x-2)\right) + 3$ to predict $f(3)$. Is $x = 3$ inside the contextual domain for this data set? Circle one: **YES** / **NO**

Explain: _____
